

Healthcare Business Problem Statement

A multi-hospital healthcare network is facing challenges related to rising treatment costs, increasing patient admissions, longer hospital stays, and inconsistent patient outcomes across hospitals and medical conditions. **The management wants to analyze historical patient data to identify:**

Which medical conditions generate the highest treatment costs
Factors contributing to longer hospital stays
Hospitals or doctors with unusually high billing amounts
Admission types causing operational pressure (Emergency, Elective, Routine)
Insurance providers associated with higher claim costs
Patterns in abnormal or inconclusive test results
Patient demographics most associated with severe conditions
The objective is to improve: Hospital resource allocation
Cost optimization
Patient care efficiency
Insurance claim management
Treatment effectiveness
Operational planning

Example Business Questions

Patient & Disease Analysis Which medical conditions are most common? Which age groups are most affected by critical diseases? Is there any gender-based trend in diseases? **Cost & Revenue**

Analysis Which medical conditions have the highest average billing amount? Which hospitals generate the highest healthcare revenue? Which insurance providers cover the most expensive treatments? **Hospital Operations Analysis** Which admission type leads to the longest hospital stay? Which hospitals experience the highest emergency admissions? What factors increase patient length of stay? **Quality of Care Analysis** Which medications are associated with better test results? Which conditions frequently return abnormal test results? Are some hospitals showing higher inconclusive test outcomes? **Business Objective** Reduce unnecessary treatment costs
Improve patient recovery outcomes
Optimize hospital occupancy and staffing
Improve emergency response planning
Negotiate better insurance partnerships
Enhance operational efficiency across hospitals